

1. Executive Summary

		Traffic Summary		Count			
Metric		Value		Total Flows Observed		31200	
Total Threats Mitigated		16		Total Packets Analyzed		2446575044	
High Priority		8		Malicious Packets		2442317620	
Low Priority		8		True Negatives Passed		22949	
Manual Releases		0					
Manual Blocks		0					
				Attack Vector		Count	
				ICMP Flood		0	
ML Metrics		Value		SYN Flood		0	
False Positives		85		UDP Flood		11	
FP Rate		0.31%		Uncertain		5	
IF Precision		99.14%					
IF Recall		94.50%		Action		Count	
IF F1-Score		96.76%		Quarantined		3	
IF Accuracy		98.25%		Time Ban		3	
				Blackhole		0	
				Released		4	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.14%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	94.50%	Share of actual attacks successfully flagged
F1-Score	96.76%	Balanced measure combining Precision and Recall
Accuracy	98.25%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.31%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	5.50%	Actual attacks that went undetected
True Positive Rate (TPR)	94.50%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.69%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 8150	FN 474
Actual: Normal	FP 71	TN 22475

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	8150

2c. Response Latency

Metric	Value	Description
Detection Time	442.3 ms	Average time taken to detect an attack
Mitigation Response Time	5.8 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.38%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	2.04%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	3.13%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.21	3	3	2026-09-16 12:25:35	2026-09-16 12:26:56
10.0.0.20	2	2	2026-09-16 12:25:34	2026-09-16 12:26:44
10.0.0.22	2	2	2026-09-16 12:25:35	2026-09-16 12:26:45
10.0.0.1	2	1	2026-09-16 12:26:27	2026-09-16 12:26:48
10.0.0.2	1	1	2026-09-16 12:26:41	2026-09-16 12:26:52

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-16 12:19:40	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-16 12:19:42	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-16 12:19:42	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-16 12:19:51	10.0.0.20	DDoS	UDP Flood	76.7%	High	Time Ban (1m)
2026-09-16 12:19:52	10.0.0.20	DDoS	UDP Flood	76.7%	High	Time Ban
2026-09-16 12:19:52	10.0.0.22	DDoS	UDP Flood	77.7%	High	Time Ban (1m)
2026-09-16 12:19:52	10.0.0.21	DDoS	UDP Flood	80.1%	High	Time Ban (1m)
2026-09-16 12:19:52	10.0.0.22	DDoS	UDP Flood	77.7%	Critical	Time Ban
2026-09-16 12:19:52	10.0.0.21	DDoS	UDP Flood	80.1%	Critical	Time Ban
2026-09-16 12:20:52	10.0.0.20	DDoS	UDP Flood	77.6%	High	Time Ban (2m)
2026-09-16 12:20:53	10.0.0.22	DDoS	UDP Flood	85.4%	High	Released
2026-09-16 12:20:53	10.0.0.21	DDoS	UDP Flood	83.7%	High	Time Ban (2m)
2026-09-16 12:21:04	10.0.0.22	DDoS	UDP Flood	85.4%	Critical	Time Ban (2m)
2026-09-16 12:26:37	10.0.0.1	DDoS	Uncertain	58.9%	Low	Released
2026-09-16 12:26:45	10.0.0.21	DDoS	UDP Flood	86.9%	High	Released
2026-09-16 12:26:52	10.0.0.2	DDoS	Uncertain	60.1%	Low	Released

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.21	UDP Flood	0.6454	86.9%	Critical	Phase 2	3	10s	2026-09-16 12:26:56	Attack persisted - escalating to time ban
10.0.0.2	Uncertain	0.5904	60.1%	Low	Phase 1	1	11s	2026-09-16 12:26:52	Attack Stopped
10.0.0.1	Uncertain	0.6090	58.9%	Low	Phase 1	1	11s	2026-09-16 12:26:48	Attack Stopped
10.0.0.22	UDP Flood	0.6199	77.4%	High	Phase 2	2	1m 10s	2026-09-16 12:26:45	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.5944	86.9%	High	Phase 2	1	1m 10s	2026-09-16 12:26:45	Ban Expired
10.0.0.20	UDP Flood	0.6389	83.7%	High	Phase 2	2	1m 10s	2026-09-16 12:26:44	Attack persisted - escalating to time ban
10.0.0.1	Uncertain	0.6117	58.9%	Low	Phase 1	1	10s	2026-09-16 12:26:37	Attack Stopped
10.0.0.22	UDP Flood	0.6266	77.4%	High	Phase 2	1	10s	2026-09-16 12:25:45	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6327	86.9%	High	Phase 2	1	10s	2026-09-16 12:25:45	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.20	UDP Flood	0.6448	75.7%	High	Phase 2	1	10s	2026-09-16 12:25:44	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE